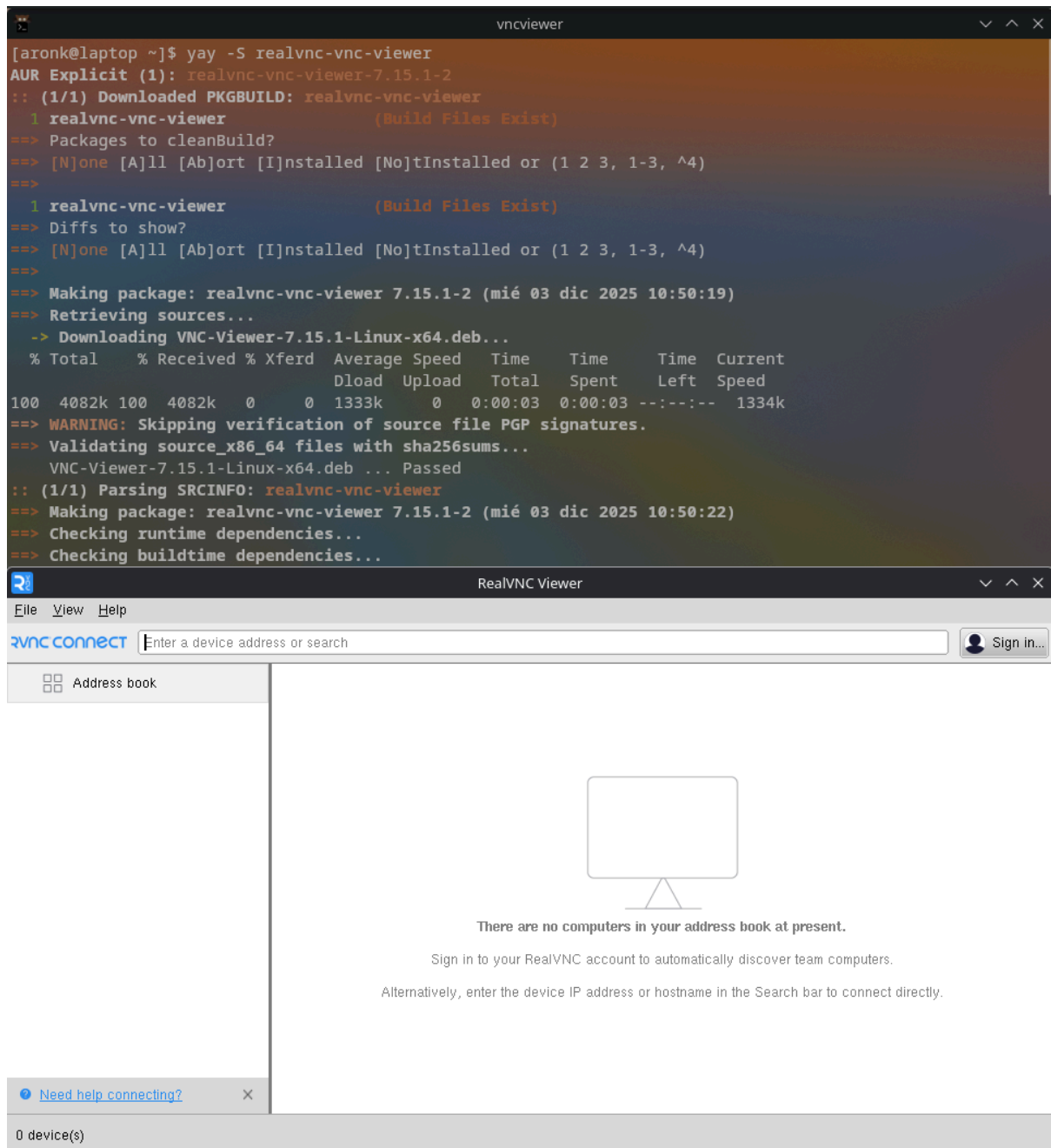


S.O linux VNC



En mi caso, tuve que instalar vnc viewer, ya que era la única opción que funcionaba actualmente para mi versión de linux. La instale mediante el comando mostrado en la primera captura y configure la conexión dentro de la propia aplicación.

```

Linux
[aronk@laptop dockers]$ cd Linux
[aronk@laptop Linux]$ docker-compose up -d
bash: docker-compose: command not found
[aronk@laptop Linux]$ docker compose up -d
WARN[0000] /home/aronk/repositorios/dockers/Linux/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 14.0s (5/11)
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 601B 0.0s
=> [internal] load build definition from Dockerfile 0.2s
=> => transferring dockerfile: 1.22kB 0.0s
=> [internal] load metadata for docker.io/consol/ubuntu-xfce-vnc:latest 2.7s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/consol/ubuntu-xfce-vnc:latest@sha256:34825e1b94fa7e438d 11.0s
=> => resolve docker.io/consol/ubuntu-xfce-vnc:latest@sha256:34825e1b94fa7e438d4 0.0s
=> => sha256:3ea050c3cbec60358ed1c9b7754b3a30fbd924c623bf7a665fe 2.14MB / 2.14MB 1.3s
=> => sha256:58f14ba6373817a8ae0f68854bbb44990b6726f05290043e67b 2.32kB / 2.32kB 0.9s
=> => sha256:6ee631c3760c35d48c1821086f7a2409e9f1043724faf1a4c8 2.10MB / 3.51MB 10.0s
=> => sha256:e6c0012b0c4b65ed3eab32c30d65ad6364ba21827f8747a 304.88kB / 304.88kB 1.8s
=> => sha256:302acf4f59837e7d73ed69773c78ea8c2e2101cec2ed76ce49 4.19MB / 87.34MB 9.2s
=> => sha256:248abfb784e825d234962385d5cf0367216302d754a772ade 4.19MB / 200.48MB 8.8s

```

```

~/repositorios/dockers/Linux
=> => extracting sha256:58f14ba6373817a8ae0f68854bbb44990b6726f05290043e67b26a9f 0.0s
=> => extracting sha256:3ea050c3cbec60358ed1c9b7754b3a30fbd924c623bf7a665fe3014f 0.1s
=> [internal] load build context 0.2s
=> => transferring context: 909B 0.0s
=> [2/6] RUN apt-get update && apt-get install -y sudo dos2unix 12.4s
=> [3/6] RUN useradd -m -s /bin/bash -G sudo DAMDAW 0.6s
=> [4/6] RUN echo 'DAMDAW:12345DAMDAW2526' | chpasswd 0.3s
=> [5/6] COPY entrypoint.sh /usr/local/bin/entrypoint.sh 0.1s
=> [6/6] RUN dos2unix /usr/local/bin/entrypoint.sh && chmod +x /usr/local/bin/en 0.3s
=> exporting to image 1.3s
=> => exporting layers 0.8s
=> => exporting manifest sha256:bdac074794f5679872e02c531fae0f300368f7ccda8dd412 0.0s
=> => exporting config sha256:b42348467877fd79ff619f5618e85ad05c5fdb8bbe9f2cc166 0.0s
=> => exporting attestation manifest sha256:53ebdbf8791faf4ffffbf8f9334a5b43c413 0.1s
=> => exporting manifest list sha256:ce6d803160c32d2c8e74bfb937bba3afed924660f28 0.0s
=> => naming to docker.io/library/linux-custom-linux-gui:latest 0.0s
=> => unpacking to docker.io/library/linux-custom-linux-gui:latest 0.2s
=> resolving provenance for metadata file 0.0s
[+] Running 3/3
✓ linux-custom-linux-gui Built 0.0s
✓ Network linux_default Created 0.0s
✓ Container linux_custom_user Started 0.4s
[aronk@laptop Linux]$

```

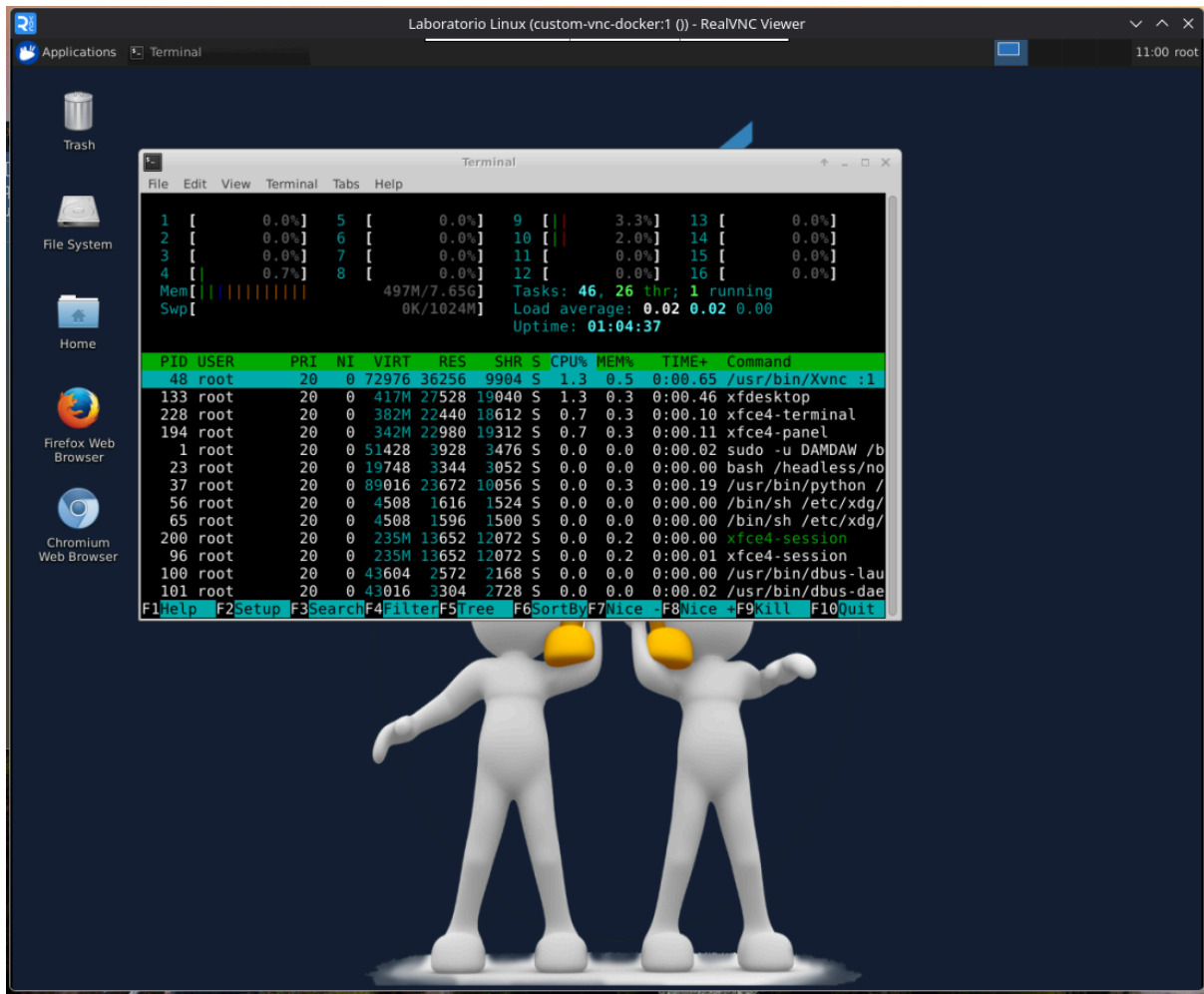
```

~/repositorios/dockers/Linux
[aronk@laptop Linux]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                     CREATED        STATUS
PORTS         NAMES
e14b5b1bc18b   linux-custom-linux-gui              "/usr/local/bin/entr..."               37 seconds ago Up 36
seconds       0.0.0.0:5901->5901/tcp, [::]:5901->5901/tcp linux_custom_user
[aronk@laptop Linux]$

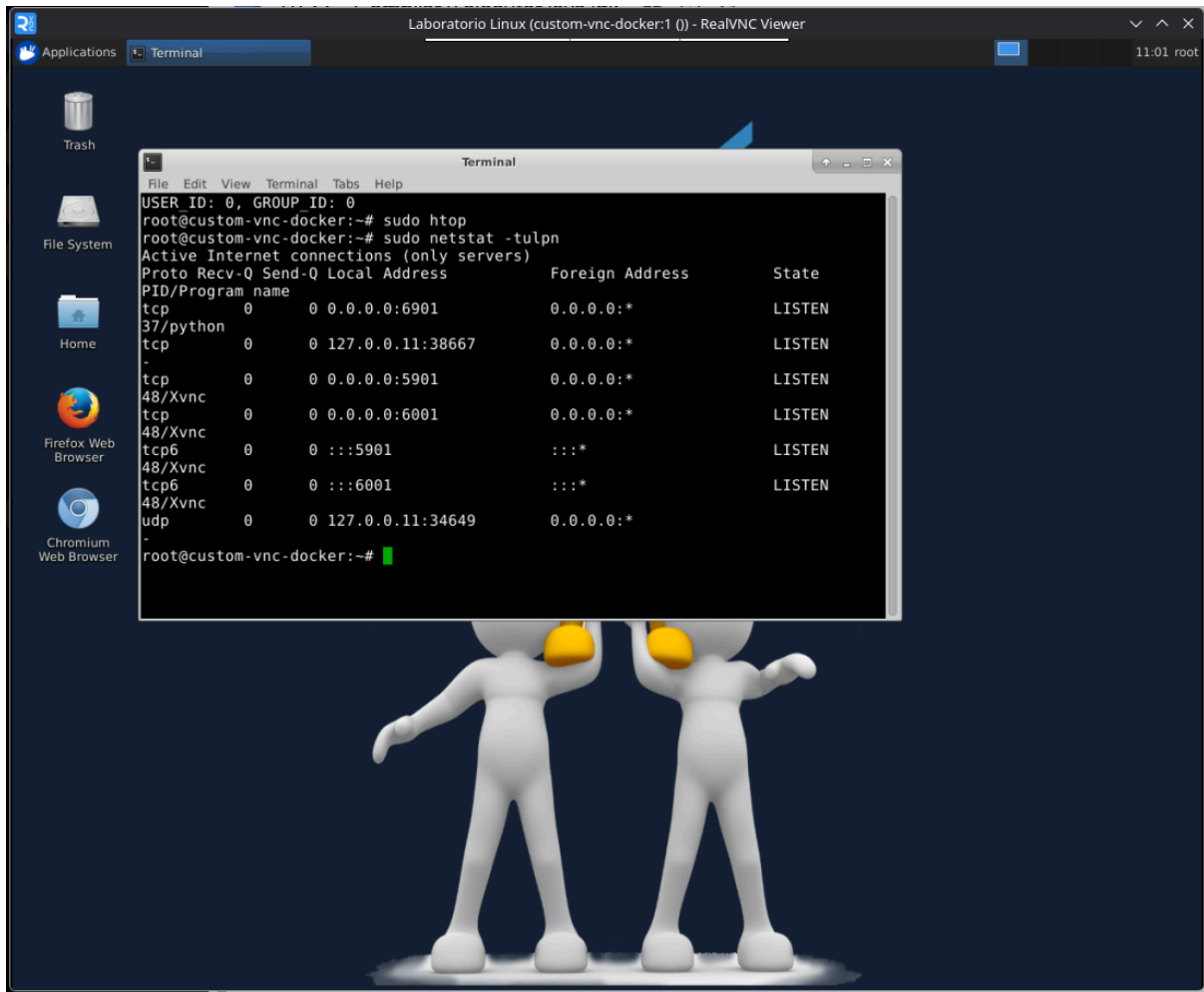
```



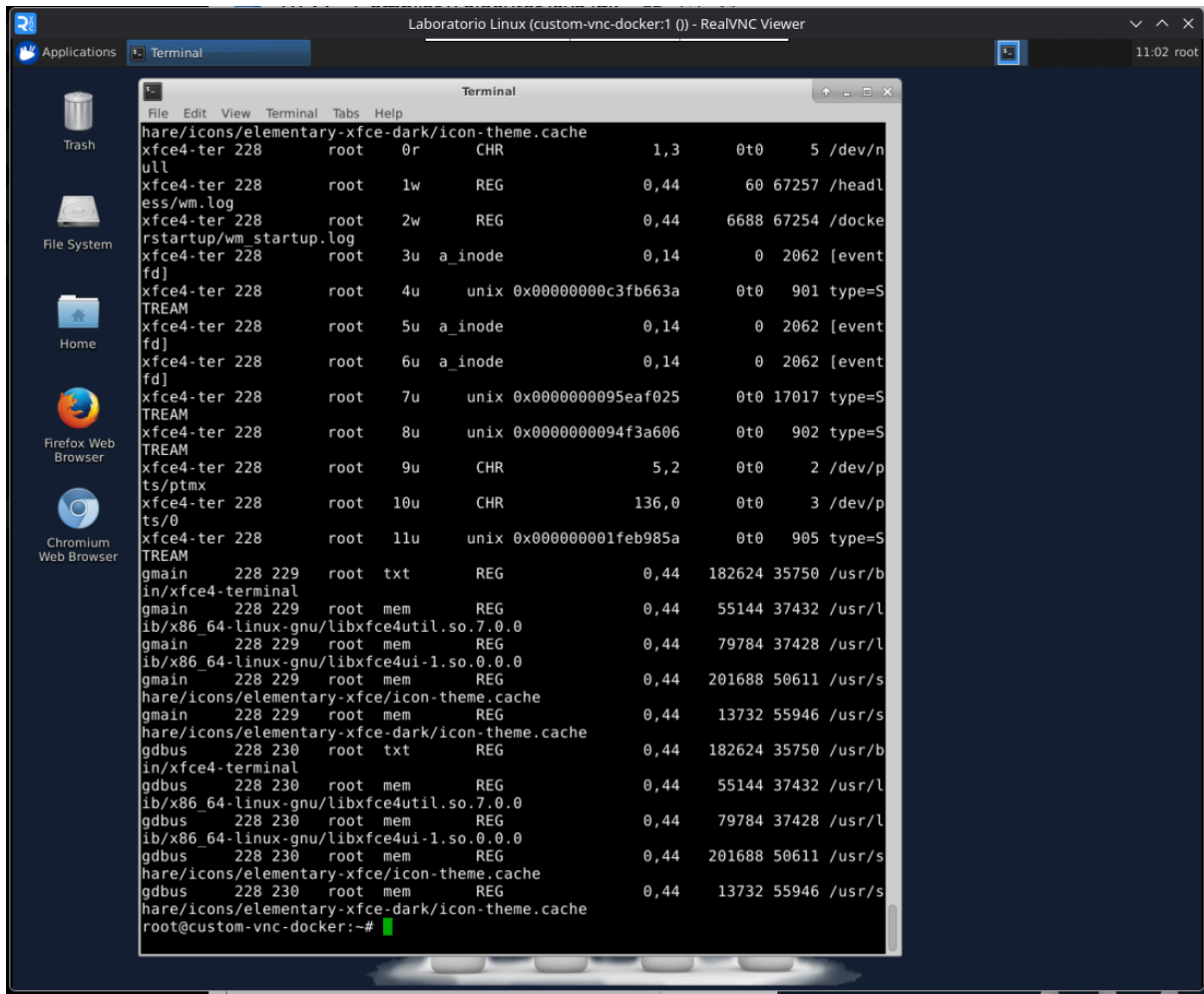
En esta captura se puede apreciar como después de haber encendido la docker con el comando 'docker compose up -d' y mediante la aplicación vnc anteriormente mencionada, he sido capaz de entrar a la docker y ver su interfaz gráfica.

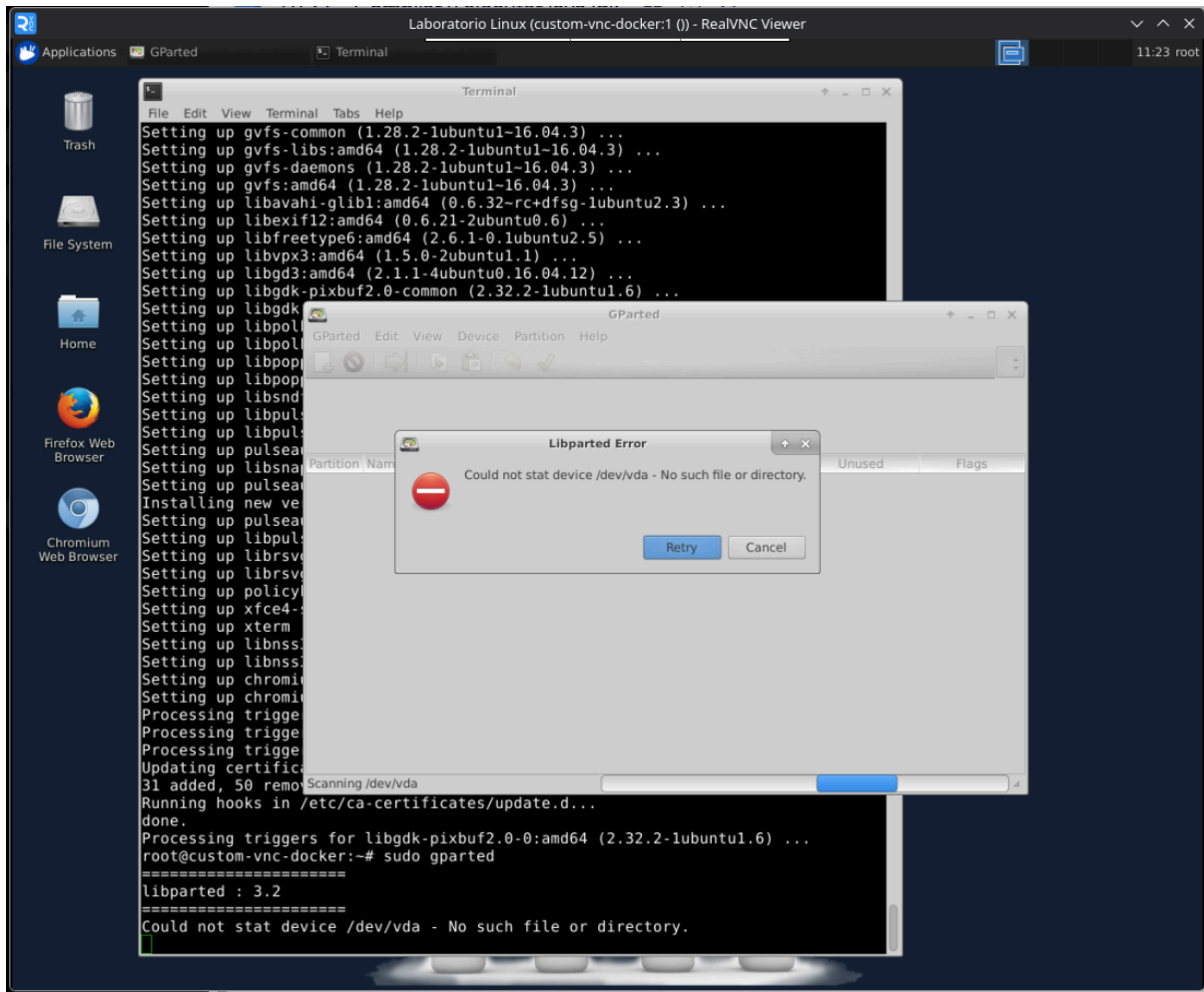


Captura que muestra el administrador de procesos dentro de la docker



Captura que muestra el estado de los puertos de la docker

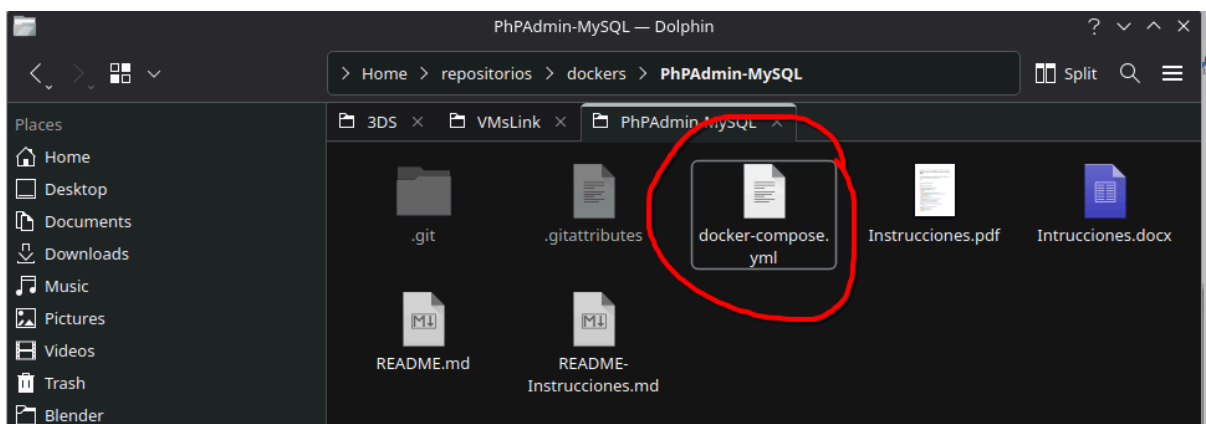




Captura que muestra la ejecución de la aplicación GParted en la docker, se puede observar que la aplicación daba un error al iniciar, esto es debido a que intentaba buscar un disco duro inexistente ya que al tratarse de una docker, no tiene un disco duro virtual como los de una máquina virtual completa, lo que causa errores que tras investigar un tiempo no he sido capaz de solucionar.

Entornos de BBDD con PHPAdmin

```
~/repositorios/dockers/PhpAdmin-MySQL
[aronk@laptop ~]$ cd repositorios/
[aronk@laptop repositorios]$ cd doc
bash: cd: doc: No such file or directory
[aronk@laptop repositorios]$ cd dockers/
[aronk@laptop dockers]$ ls
Linux
[aronk@laptop dockers]$ git clone https://github.com/jrmcero07/PhpAdmin-MySQL.git
Cloning into 'PhpAdmin-MySQL'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 18 (delta 4), reused 12 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 6.00 MiB | 2.57 MiB/s, done.
Resolving deltas: 100% (4/4), done.
[aronk@laptop dockers]$ ls
Linux  PhpAdmin-MySQL
[aronk@laptop dockers]$ cd PhpAdmin-MySQL/
[aronk@laptop PhpAdmin-MySQL]$ ls
docker-compose.yml  Instrucciones.pdf  Instrucciones.docx  README-Instrucciones.md  README.md
[aronk@laptop PhpAdmin-MySQL]$
```



Tras clonar el repositorio para la docker que son una base de datos, podemos comprobar que en el archivo docker-compose.yml estaba ya preparado para la correcta realización de este apartado.


```

~/repositorios/dockers/PhpAdmin-MySQL
[aronk@laptop PhpAdmin-MySQL]$ docker compose up -d
WARN[0000] /home/aronk/repositorios/dockers/PhpAdmin-MySQL/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/36
  ✓ phpmyadmin Pulled                20.4s
  ✓ db Pulled                        16.6s

[+] Running 4/4
  ✓ Network phpadmin-mysql_default C... 0.1s
  ✓ Volume phpadmin-mysql_db_data Cr... 0.0s
  ✓ Container mysql-dev-bbdd Started 0.8s
  ✓ Container phpmyadmin-dev-gui Sta... 0.6s
[aronk@laptop PhpAdmin-MySQL]$

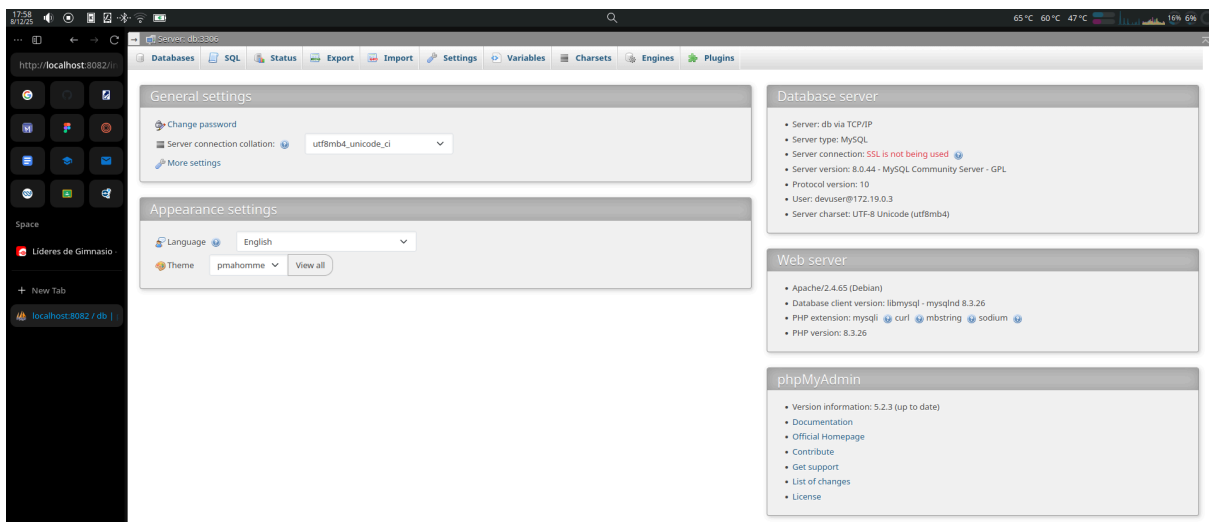
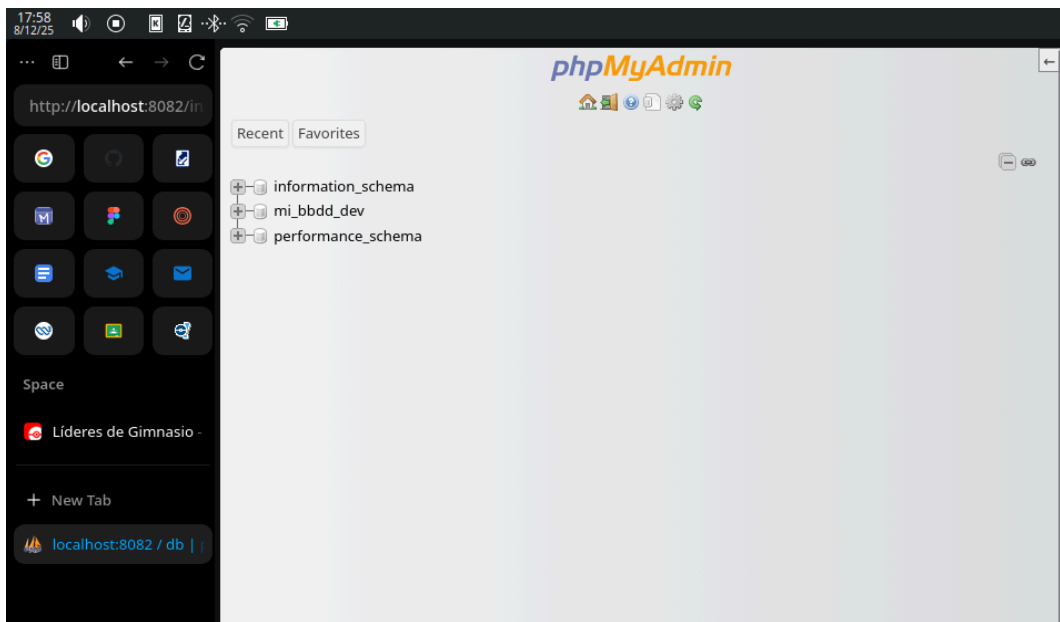
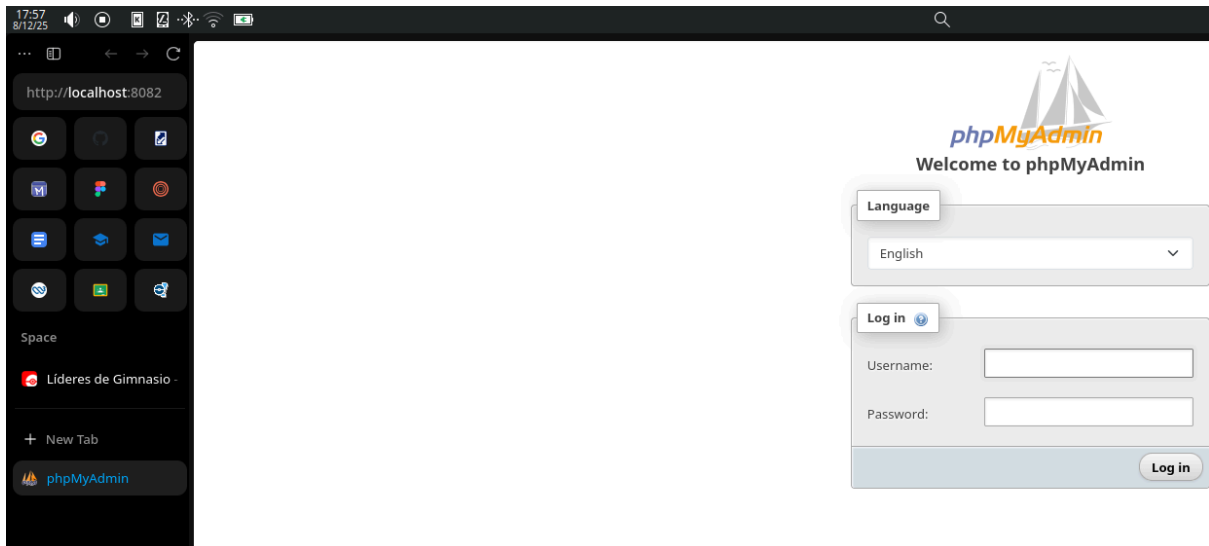
```

```

~/repositorios/dockers/PhpAdmin-MySQL
[aronk@laptop PhpAdmin-MySQL]$ docker compose ps
WARN[0000] /home/aronk/repositorios/dockers/PhpAdmin-MySQL/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
mysql-dev-bbdd      mysql:8.0            "docker-entrypoint.s..." db          2 minutes ago  Up 2 minutes  0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
phpmyadmin-dev-gui  phpmyadmin/phpmyadmin "/docker-entrypoint..." phpmyadmin  2 minutes ago  Up 2 minutes  0.0.0.0:8082->8082/tcp, [::]:8082->8082/tcp
[aronk@laptop PhpAdmin-MySQL]$

```

Capturas que demuestran el correcto arranque de la docker correspondiente.



Capturas que demuestran el correcto funcionamiento del servicio de bbdd conectandome mediante el navegador a la docker.

```
docker exec -it mysql-dev-bbdd bash

[aronk@laptop PHPAdmin-MySQL]$ docker exec -it mysql-dev-bbdd bash
bash-5.1# mysql -u devuser -p mi_bbdd_dev
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 24
Server version: 8.0.44 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

```
mysql>
mysql>
mysql>
mysql> SQL CREATE TABLE productos (
-> id INT AUTO:INCREMENT PRIMARY KEY,
-> nombre VARCHAR(100) NOT NULL,
-> precio DECIMAL(10,2)
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corre
sponds to your MySQL server version for the right syntax to use near 'SQL CREATE TABL
E productos (
id INT AUTO:INCREMENT PRIMARY KEY,
nombre VARCHAR(1' at line 1
mysql> CREATE TABLE productos (id INT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(100)
NOT NULL, precio DECIMAL(10,2) );
Query OK, 0 rows affected (0.09 sec)

mysql>
```

```
mysql> INSERT INTO productos (nombre, precio) VALUES ('Laptop Dev', 1200.50);
Query OK, 1 row affected (0.04 sec)

mysql>
```

```
mysql> SELECT * FROM productos;
+-----+-----+-----+
| id | nombre      | precio |
+-----+-----+-----+
|  1 | Laptop Dev  | 1200.50 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

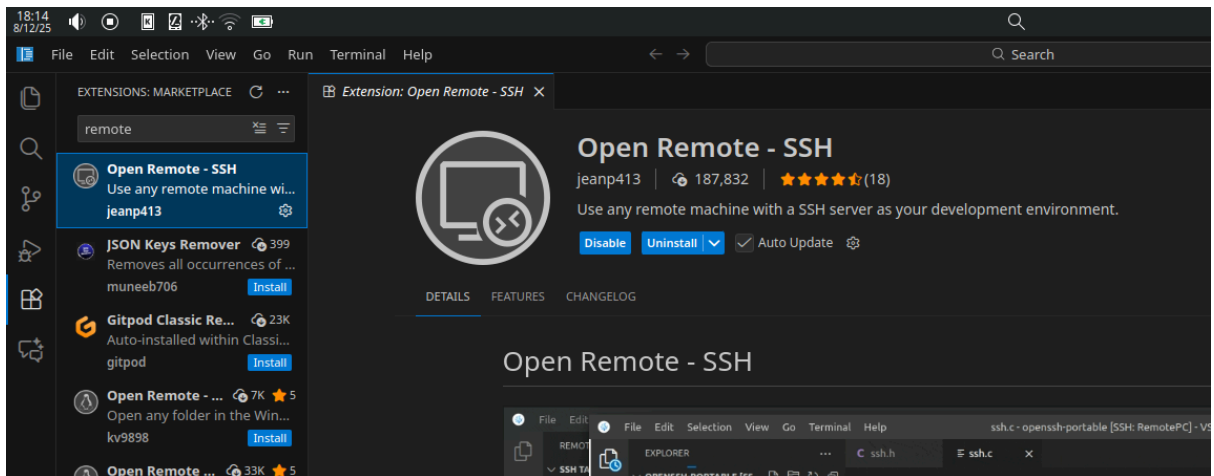
Capturas que demuestran el correcto funcionamiento de los comandos de administración de mysql en la docker creada.

```
bash-5.1# exit
exit
[aronk@laptop PHPAdmin-MySQL]$ docker compose stop
WARN[0000] /home/aronk/repositorios/dockers/PHPAdmin-MySQL/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Stopping 2/2
  ✓ Container phpmyadmin-dev-gui   Stopped      1.3s
  ✓ Container mysql-dev-bbdd       Stopped      1.0s
[aronk@laptop PHPAdmin-MySQL]$
```

Entornos de programación

```
~/repositorios/dockers
[aronk@laptop dockers]$ git clone https://github.com/jrmcero07/Python-VSCode.git
Cloning into 'Python-VSCode'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 12 (delta 0), reused 12 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), done.
[aronk@laptop dockers]$
```

Captura de clonación del repositorio preparado con un entorno de programación.



```

~/repositorios/dockers/Python-VSCode
[aronk@laptop Python-VSCode]$ docker compose up -d
WARN[0000] /home/aronk/repositorios/dockers/Python-VSCode/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 7/7
  ✓ python-dev Pulled 8.3s
    ✓ 1771569cc129 Pull complete 0.6s
    ✓ 0e4bc2bd6656 Pull complete 2.4s
    ✓ 22b63e76fde1 Pull complete 2.5s
    ✓ b3dd773c3296 Pull complete 2.8s
    ✓ 3b50134712e6 Download complete 0.0s
    ✓ d927adb2c82a Download complete 0.0s
[+] Running 2/2
  ✓ Network python-vscode_dev-network Cr... 0.0s
  ✓ Container python_emv_dev Started 0.4s
[aronk@laptop Python-VSCode]$

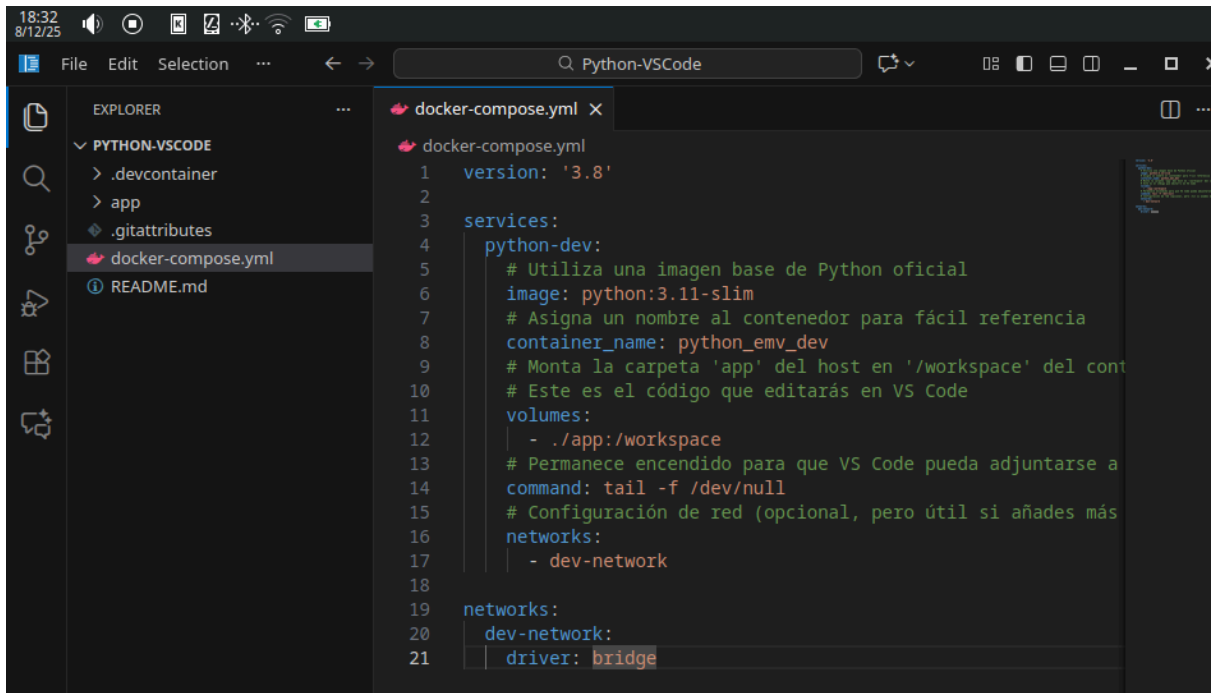
```

```

~/repositorios/dockers/Python-VSCode
[aronk@laptop Python-VSCode]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS
77c94f1a9e5f   python:3.11-slim                    "tail -f /dev/null"                    2 minutes ago  Up
2 minutes                                           python_emv_dev
fa5000796e7a   phpmyadmin/phpmyadmin              "/docker-entrypoint..."              36 minutes ago Up
2 minutes  0.0.0.0:8082->80/tcp, [::]:8082->80/tcp  phpmyadmin-dev-gui
85bf05dc59d6   mysql:8.0                          "docker-entrypoint.s..."             36 minutes ago Up
2 minutes  0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp  mysql-dev-bbdd
e14b5b1bc18b   linux-custom-linux-gui             "/usr/local/bin/entr..."             5 days ago    Up
2 minutes  0.0.0.0:5901->5901/tcp, [::]:5901->5901/tcp  linux_custom_user
[aronk@laptop Python-VSCode]$

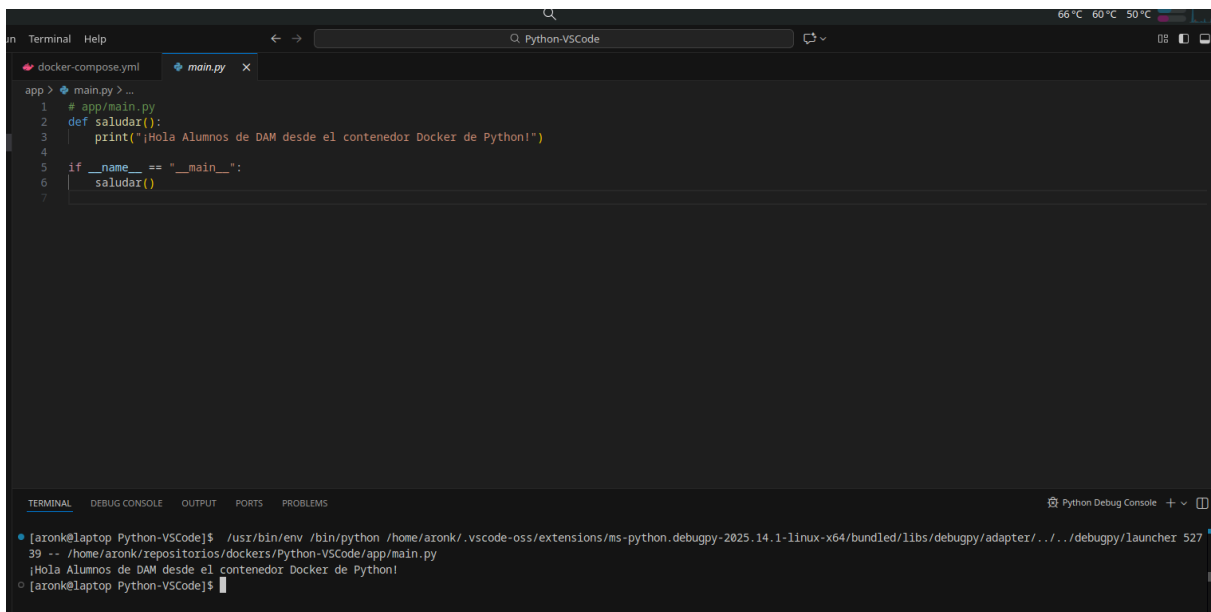
```

Capturas que demuestra la correcta instalación de la docker de este apartado.



The screenshot shows the Visual Studio Code interface with the 'EXPLORER' sidebar on the left. The 'PYTHON-VSCODE' folder is expanded, showing subfolders like '.devcontainer' and 'app', and files like '.gitattributes', 'docker-compose.yml', and 'README.md'. The 'docker-compose.yml' file is open in the editor, displaying the following content:

```
1 version: '3.8'
2
3 services:
4   python-dev:
5     # Utiliza una imagen base de Python oficial
6     image: python:3.11-slim
7     # Asigna un nombre al contenedor para fácil referencia
8     container_name: python_emv_dev
9     # Monta la carpeta 'app' del host en '/workspace' del cont
10    # Este es el código que editarás en VS Code
11    volumes:
12      - ./app:/workspace
13    # Permanece encendido para que VS Code pueda adjuntarse a
14    command: tail -f /dev/null
15    # Configuración de red (opcional, pero útil si añades más
16    networks:
17      - dev-network
18
19 networks:
20   dev-network:
21     driver: bridge
```



The screenshot shows the Visual Studio Code interface with the 'main.py' file open in the editor. The file content is as follows:

```
1 # app/main.py
2 def saludar():
3     print("¡Hola Alumnos de DAM desde el contenedor Docker de Python!")
4
5 if __name__ == "__main__":
6     saludar()
7
```

Below the editor, the 'TERMINAL' panel is visible, showing the command prompt and the output of the program:

```
[aronk@laptop Python-VSCODE]$ /usr/bin/env /bin/python /home/aronk/.vscode-oss/extensions/ms-python.debugpy-2025.14.1-linux-x64/bundled/libs/debugpy/adapter/.../debugpy/launcher 527
39 -- /home/aronk/repositorios/dockers/Python-VSCODE/app/main.py
¡Hola Alumnos de DAM desde el contenedor Docker de Python!
[aronk@laptop Python-VSCODE]$
```

Capturas que demuestran la correcta realización de los apartados pedidos en esta docker.